XII. Maximum Likelihood Estimation (GSE 518)

Carlos Flores
Department of Economics
California Polytechnic State University*

In these notes we discuss the second estimation approach we cover in this class: *Maximum Likelihood Estimation.*

1. Motivation

- Let X_i be discrete and assume $f_X(x|\theta_0) = Pr(X = x|\theta_0)$ for some $\theta_0 \in \Theta$.
  - Example: Let $X \sim Bernoulli(p_0)$.
- MLE asks: What is the probability of observing the *particular* random sample we got $(X_1 = x_1, X_2 = x_2 \ldots, X_n = x_n)$ for *any* given $\theta \in \Theta$?
  - Example (Continuation): (See details in lecture)
- **MLE idea:** take the sample $(X_1 = x_1, X_2 = x_2 \ldots, X_n = x_n)$ as *fixed* and choose the value of θ that is *most likely* to have generated our sample.
  - In other words, what is the value of θ that maximizes the chances of getting the *specific* sample we have?
  - Example (Continuation): (See details in lecture)
  - The same intuition applies to the case when X is a continuous random variable. (See example from lecture.)

*MSQE Alumni Hunter Happersett contributed significantly to writing these lecture notes in Latex.

2. Maximum Likelihood Estimator (MLE): Definition

Definition: The *likelihood function* of θ for an iid sample $(X_1, \ldots, X_n) \sim f(x|\theta_0)$ is

$$\ell(\theta|x_1, \ldots, x_n) = \prod_{i=1}^{n} f(X_i|\theta) = f(X_1, \ldots, X_n|\theta)$$

Notes on likelihood function:

- In the discrete case, $\ell(\theta|x_1, \ldots, x_n)$ gives the probability of getting the data we have.
  - * The same intuition applies for the continuous case.
- In $\ell(\theta|x_1, \ldots, x_n)$, we think of $x_1, \ldots, x_n$ as fixed (hence the lowercase letters) and allow θ to vary $\theta \in \Theta$.

We typically work with the *log-likelihood* function:

$$L(\theta|x_1, \ldots, x_n) = \ln\left(\ell(\theta|x_1, \ldots, x_n)\right) = \sum_{i=1}^{n} \ln\left(f(x_i|\theta)\right)$$

Definition: The *Maximum Likelihood Estimator (MLE)* $\widehat{\theta}_{MLE}$ of θ_0 is the value of $\theta \in \Theta$ that maximizes the log-likelihood $L(\theta|x_1, \ldots, x_n)$. That is,

$$\widehat{\theta}_{MLE} = \arg\max_{\theta \in \Theta} L(\theta|x_1, \ldots, x_n) = \arg\max_{\theta \in \Theta} \sum_{i=1}^{n} \ln\left(f(x_i|\theta)\right)$$

Notes on the MLE Definition:

- Indeed, note that the MLE estimator *maximizes* the likelihood (or probability) of observing the sample we have.

- Steps to obtain $\widehat{\theta}_{MLE}$:

  1. Get $\ln\left(f(x_i|\theta)\right)$
  2. Get $\sum_{i=1}^{n} \ln\left(f(x_i|\theta)\right)$
  3. Get $\arg\max_{\theta\in\Theta} \sum_{i=1}^{n} \ln\left(f(x_i|\theta)\right)$, i.e., maximize $\sum_{i=1}^{n} \ln\left(f(x_i|\theta)\right)$ with respect to θ.

- Example 1: Let $(X_1, \ldots, X_n) \overset{\text{iid}}{\sim} Bernoulli(p_0)$. Find $\widehat{p}_{MLE}$. Take as given that the pmf of X is:

$$f(x|p_0) = p_0^x (1-p_0)^{1-x} \quad \text{for } x \in \{0, 1\}$$

  1. Compute the log of the pmf for one observation (i.e., one X_i):

$$\ln(f(x \mid p)) = \ln(p^x (1-p)^{1-x}) = x \ln(p) + (1-x)\ln(1-p)$$

  2. Sum the log-likelihoods of all observations:

$$L(p \mid \cdot) = \sum_{i=1}^{n} \ln(f(x_i \mid p)) = \sum_{i=1}^{n} \left(x_i \ln(p) + (1-x_i)\ln(1-p)\right)$$

  3. Maximize the log-likelihood $L(p \mid \cdot)$ with respect to p:

$$\frac{\partial L(p \mid \cdot)}{\partial p} = \sum_{i=1}^{n} \frac{x_i}{p} + \sum_{i=1}^{n} \frac{(1-x_i)(-1)}{1-p} = 0$$

$$\sum_{i=1}^{n} \frac{x_i}{p} + \frac{1}{1-p}\left(n - \sum_{i=1}^{n} x_i\right) = 0$$

$$\sum_{i=1}^{n} x_i = p\left(n - \sum_{i=1}^{n} x_i\right)$$

$$\sum_{i=1}^{n} x_i = np - p\sum_{i=1}^{n} x_i$$

$$\boxed{\widehat{p}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}}$$

(Note: We write capital X in the last line to stress that, even though MLE "thinks" of the sample as fixed, the estimator is indeed a function of the random sample $(X_1, X_2, \ldots, X_n)$ and is hence random.)

- Example 2: Let $(X_1, \ldots, X_n) \overset{iid}{\sim} Normal(\mu_0, \sigma_0^2)$. Find the MLE estimators for μ_0 and σ_0^2.

  (See solution from lecture.)

3. Notes on MLE

1. Sometimes $\widehat{\theta}_{\text{MOM}} = \widehat{\theta}_{MLE}$ (as in our example with the normal distribution above) but this is not always the case.

2. Drawbacks:

   (i) $\widehat{\theta}_{MLE}$ may not have a closed-form solution.

   * In this case, we use *numerical methods* to maximize the log-likelihood $L(\theta \mid \cdot)$ (e.g., probit and logit models). This is also true for MOM (and GMM) methods.

   (ii) $L(\theta \mid \cdot)$ may not be strictly concave.

   * It can be hard to find a global maximum in such cases.

   * Recall that calculus can only find local maxima.

(iii) It is a "very" parametric approach (we are assuming a lot):

* We assume $X \sim f(\cdot \mid \theta_0)$ is known.
* If $X \not\sim f(\cdot \mid \theta_0)$, conclusions are wrong.

(iv) For many (most?) models, $\widehat{\theta}_{MLE}$ is biased in finite samples (i.e., its expected value is different from θ_0, so its distribution is *not* centered at θ_0) (later).

3. $\widehat{\theta}_{MLE}$ has very nice asymptotic properties (i.e., as $n \to \infty$) (to be covered later):

- Consistency: $\widehat{\theta}_{MLE} \xrightarrow{P} \theta_0$
- Asymptotically normally distributed: $\widehat{\theta}_{MLE} \sim N(\theta_0, \Omega)$ for some variance Ω
- Asymptotically efficient.

4. Invariance Property of MLE: (*Thm. 7.2.10. in textbook.*) If $\widehat{\theta}_{MLE}$ is the MLE of θ_0, then for any function $\gamma_0 = g(\theta_0)$, the MLE of γ_0 is $\widehat{\gamma}_{MLE} = g(\widehat{\theta}_{MLE})$.

- In words, to get the MLE estimator of a function of our parameter, say $g(\theta_0)$, we just plug our estimator $\widehat{\theta}_{MLE}$ of θ_0 in the function: $g(\widehat{\theta}_{MLE})$.
- Used a lot in econometrics.
- Example: Let $(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu_0, \sigma_0^2)$.
  - Above, we found the MLE estimator of the variance: $\widehat{\sigma}^2_{MLE} = \widetilde{s}^2$
  - Suppose we now want to find the MLE of the standard deviation $\sigma_0 = g(\sigma_0^2) = \sqrt{\sigma_0^2}$
  - By the theorem above, $\widehat{\sigma}_{MLE} = \sqrt{\widehat{\sigma}^2_{MLE}} = \sqrt{\widetilde{s}^2} = \widetilde{s}$.

5. Score Function (Definition): The *score function* is given by

$$S(\theta, x) = \frac{\partial \ln f(x \mid \theta)}{\partial \theta}$$

- It is the derivative of the log-likelihood with respect to the parameter (θ) for *only one observation* (no $\sum_{i=1}^{n} x_i$).

Score Identity (Result): $E[S(\theta_0, X)] = 0$

- The expected value of the Score function *when evaluated at true value of* θ must be zero.
- Remember: θ_0 is the true value of $\theta \in \Theta$, θ is *any* value $\theta \in \Theta$, and $\widehat{\theta}$ is the estimator.
- Rough sketch of proof: Start with the fact that, for any $\theta \in \Theta$, $f(x|\theta)$ is a pdf and hence $\int_{\infty}^{\infty} f(x|\theta)dx = 1$. Take the derivative wrt θ to each side of the equality. After some algebra, and by noting that the *true* pdf of X is $f(x|\theta_0)$, we get the result. This last part ensures the score identity holds when evaluated at θ_0.
- *Intuition/Analogy:* Like the first order condition (FOC) for maximization.
  - Let $y = f(x)$ and suppose $f(x)$ has a maximum at x^*. Then, at x^* we know that $dy/dx = f'(x^*) = 0$.
  - The Score Identity is analogous to the FOC to find the maximum, since it can be shown that θ_0 maximizies $E[ln(f(x|\theta)]$ over $\theta \in \Theta$ (later).

Example: Let $(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu_0, 1)$. Find the score function and check the score identity.

- Given $X \sim N(\mu_0, 1)$, we know the pdf of X is given by:
  $f(x; 1, \mu) = (2\pi)^{-1/2} e^{-(x-\mu)^2/2}$
- *Find the log of the pdf:* $\ln(f(x; 1, \mu)) = -\frac{1}{2}\ln(2\pi) - \frac{(x-\mu)^2}{2}$
- *Find the Score Function:* $S(\mu, X) = \frac{\partial \ln f(x|\mu)}{\partial \mu} = -\frac{1}{2} \cdot 2(X - \mu)(-1) = X - \mu$
  Hence, $\boxed{S(\mu, X) = X - \mu}$

- *Verify the Score Identity:* Given that $X_i \sim N(\mu_0, 1)$:

  $$\boxed{E[S(\mu_0, X_i)] = \mu_0 - \mu_0 = 0}$$

  Note that in general $E[S(\mu, X)] \neq 0$:

  $$E[S(\mu, X)] = E[X_i - \mu] = \mu_0 - \mu \neq 0 \text{ at any general value } \mu$$

6. Information Matrix Identity (Result): Let θ be a scalar. Under some conditions (in essence that it is possible to interchange the derivatives and integrals; see Lemma 7.3.11 & 7.3.9 in book),

$$-E\left[\frac{\partial^2 \ln f(x \mid \theta_0)}{\partial \theta^2}\right] = E\left[\left(\frac{\partial \ln f(x \mid \theta)}{\partial \theta}\right)^2\right], \quad \text{or}$$

$$-E\left[\frac{\partial S(\theta_0, X)}{\partial \theta}\right] = E\left[[S(\theta_0, X)]^2\right]$$

   - As for the score identity, everything *must be evaluated at* θ_0 *(true value of the parameter)* for the equalities to hold.
   - An example in which it is not possible to interchange the derivative and integral (i.e., $E(\cdot)$) is when the sample space of X depends on θ_0. In that case, the Score and Information Identities may not hold.
   - Rough sketch of proof: It is similar to the score identity proof, except we take the *second* derivative with respect to θ of the equality $\int_{\infty}^{\infty} f(x|\theta)dx = 1$. As before, after some algebra and by noting that the *true* pdf of X is $f(x|\theta_0)$, we get the result.
   - More generally, when the parameter is a $k \times 1$ vector, the term in each side of the equality is a $k \times k$ matrix (like a variance-covariance matrix).

Example (Continuation): As before, let $(X_1, \ldots, X_n) \overset{iid}{\sim} N(\mu_0, 1)$. Check that the Information Matrix Identity holds. Recall that above we found: $S(\mu, X) = X - \mu$.

- *Get left-hand term:* $\frac{\partial S(\mu, X)}{\partial \mu} = -1 \implies -E\left[S(\mu_0, X)\right] = 1$
- *Get right-hand term:* $E\left[S^2(\mu_0, x)\right] = E[(X - \mu_0)^2] = Var(X) = 1$, since $X \sim N(\mu_0, 1)$

  Hence, $\boxed{-E\left[S(\mu_0, X)\right] = E\left[S^2(\mu_0, X)\right] = 1}$

  Note that in general $-E\left[S(\mu_0, X)\right] \neq E\left[S^2(\mu_0, X)\right]$, unless we evaluate at θ_0, the *true* value of θ:

  $$E[S^2(\mu, X)] = E[(X - \mu)^2] \neq Var(X) = 1, \text{ unless } \mu = E(X) = \mu_0.$$

  In other words, $E[(X - \mu)^2]$ is *not* the variance of X unless it is evaluated at μ_0, the true mean of X.

7. Alternative Way to Motivate MLE:

- Result: Let $X \sim f(x \mid \theta_0)$ with $\theta_0 \in \Theta$. Then, θ_0 maximizes $E[\ln f(x, \theta)]$ with respect to $\theta \in \Theta$. That is,

  $$E[\ln f(x \mid \theta_0)] \geq E[\ln f(x \mid \theta)], \qquad \text{for all } \theta \in \Theta$$

  - See proof in lecture. The proof of the Score and Information Matrix Identities have a similar "flavor".

- MLE Idea: By the result above, we know that in the *population* it is true that

  $$\theta_0 = \arg\max_{\theta \in \Theta} E[\ln f(x \mid \theta)]$$

Hence, we estimate θ_0 by doing the same in the *sample*, using the *empirical expectation* $\frac{1}{n}\sum \cdot$ (i.e., sample means) instead of the *population expectation* $E(\cdot)$. Specifically, $\widehat{\theta}_{MLE}$ maximizes the sample average of $\ln f(x \mid \theta)$, or

$$\widehat{\theta}_{MLE} = \arg\max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^{n} \ln\left(f(x_i|\theta)\right)$$

- This general idea of finding estimators ("if something is true in the population, we do the same in the sample") is widely used. Such estimators are usually called *analog estimators*.
- *Intuition for "why MLE works":* As sample size goes to infinity ($n \to \infty$), since $\frac{1}{n}\sum_{i=1}^{n} \ln f(x_i \mid \theta)$ will be "close to" $E[\ln f(x \mid \theta)]$ (it is a sample average!) $\implies$ $\widehat{\theta}_{MLE}$ will be "close to" θ_0.
- *Intuition for Score Identity:* Recall the Score Identity: $E[S(\theta_0, X)] = 0$. Since θ_0 maximizes $E[\ln f(x \mid \theta)]$, the first order conditions imply:

$$\left.\frac{\partial E[\ln f(x \mid \theta)]}{\partial \theta}\right|_{\theta=\theta_0} = 0 \implies E\left[\left.\frac{\partial \ln f(x \mid \theta)}{\partial \theta}\right|_{\theta=\theta_0}\right] = E[S(\theta_0, X)] = 0$$

8. Many estimators in econometrics are MLE. In general, if we assume $X \sim f(x \mid \theta)$ is known, most times use MLE. Examples:

   - OLS
   - Logit/Probit, Multinomial Logit/Probit, Ordered Logit/Probit
   - Poisson regression
   - Duration/Hazard Models, ...